Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(5)$.

Final Answer: 14. Substitute $x = 5$: $f(5) = 2(5) + 4 = 14$.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 104°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 45^\circ) = 104^\circ$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 103°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor:$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56(3x)^3 = 27x^3(-2y^2)^5 = -32y^{10}$Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32)x^3y^{10} = 56 \times (-864)x^3y^{10} = -48384x^3y^{10}$.

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the sum of all integers between 10 and 100 (inclusive) that are divisible by 4.

1. Identify the sequence: The first multiple of $4 \geq 10$ is $a_1 = 12$. The last multiple ≤ 100 is $a_n = 100$. The common difference is $d = 4$. 2. Find the number of terms n:

$a_n = a_1 + (n-1)d \Rightarrow 100 = 12 + (n-1)4 \Rightarrow 88 = 4(n-1) \Rightarrow n - 1 = 22 \Rightarrow n = 23.$ 3. Calculate the sum using $S_n = \frac{n}{2}(a_1 + a_n)$:

$S_{23} = \frac{23}{2}(12 + 100) = \frac{23}{2}(112) = 23 \cdot 56 = 1288.$

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a geometric sequence, the 3rd term is 18 and the 6th term is 486. Find the common ratio r, the first term a_1, and the sum of the first 5 terms S_5.

Use the general formula $a_n = a_1 r^{n-1}$: $a_3 = a_1 r^2 = 18, \quad a_6 = a_1 r^5 = 486$. Divide a_6 by a_3: $\frac{a_1 r^5}{a_1 r^2} = r^3 = \frac{486}{18} = 27 \Rightarrow r = 3$. Solve for a_1: $a_1(3^2) = 18 \Rightarrow 9a_1 = 18 \Rightarrow a_1 = 2$. Compute S_5 using the geometric series sum formula: $S_5 = \frac{a_1(r^5 - 1)}{r - 1} = \frac{2(3^5 - 1)}{3 - 1} = \frac{2(243 - 1)}{2} = 242$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1 - r}$: $25 = \frac{a_1}{1 - (-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1 - r^4)}{1 - r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1 - 0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1 - r} = \frac{12}{1 - 2/3} = \frac{12}{1/3} = 36$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An infinite geometric series has first term $a_1 = 36$ and converges to $S_\infty = 48$. What is the common ratio r, and what is the third term a_3?

Using the infinite geometric sum formula $S_\infty = \frac{a_1}{1 - r}$: $48 = \frac{36}{1 - r} \Rightarrow 1 - r = \frac{36}{48} = \frac{3}{4} \Rightarrow r = 1 - \frac{3}{4} = \mathbf{\frac{1}{4}}$ Since $|r| = \frac{1}{4} < 1$, the convergence condition holds. Find the third term $a_3 = a_1 r^2$: $a_3 = 36\left(\frac{1}{4}\right)^2 = 36 \times \frac{1}{16} = \mathbf{\frac{9}{4}} = 2.25$

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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the finite series $\sum_{k=1}^{4}(2k-1)$.

Step 1: Expand the summation terms for $k = 1, 2, 3, 4$: $k = 1 : 2(1) - 1 = 1$ $k = 2 : 2(2) - 1 = 3$ $k = 3 : 2(3) - 1 = 5$ $k = 4 : 2(4) - 1 = 7$ Step 2: Add the terms: $1 + 3 + 5 + 7 = \mathbf{16}$ (Property: The sum of the first n odd positive integers is $n^2 = 4^2 = 16$).

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An infinite geometric series has first term $a_1 = 5$ and common ratio $r = 0.6$. Find its sum S_∞.

Since $|r| = |0.6| < 1$, the infinite geometric series converges. Apply $S_\infty = \frac{a_1}{1-r}$:
$S_\infty = \frac{5}{1-0.6} = \frac{5}{0.4} = \frac{5}{2/5} = \mathbf{\frac{25}{2} = 12.5}$

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A polynomial $f(x)$ of degree 4 with real coefficients has leading coefficient 2 and zeros at $x = -1$ (multiplicity 2) and $x = 2 + i$. Write $f(x)$ in fully expanded standard form.

Step 1: Apply the Conjugate Root Theorem: Since $f(x)$ has real coefficients, the complex conjugate $2 - i$ must also be a zero. Step 2: Construct the factored form: $f(x) = 2(x+1)^2((x-2)-i)((x-2)+i) = 2(x^2+2x+1)((x-2)^2+1)$. Step 3: Expand and simplify: $(x-2)^2 + 1 = x^2 - 4x + 5$.

$f(x) = 2(x^2+2x+1)(x^2-4x+5) = 2(x^4-2x^3-2x^2+6x+5) = \mathbf{2x^4-4x^3-4x^2+12x+10}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability & Statistics

For what values of k does $x^2 - kx + 9 = 0$ have equal real roots?

Step 1: A quadratic equation $ax^2 + bx + c = 0$ has equal real roots if and only if its discriminant equals zero: $\Delta = b^2 - 4ac = 0$. Step 2: Substitute $a = 1$, $b = -k$, and $c = 9$:
$(-k)^2 - 4(1)(9) = 0 \Rightarrow k^2 - 36 = 0 \Rightarrow k^2 = 36$ Step 3: Solve for k: $k = \pm\sqrt{36} \Rightarrow \mathbf{k = \pm 6}$

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 2.01 Quadratics & Complex Numbers**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x - 2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 3.01 Polynomial Functions**

Find the average rate of change of the polynomial function $f(x) = 2x^3 - 5x + 4$ over the closed interval $[-1, 2]$.

Step 1: Evaluate the function at the interval endpoints: $f(-1) = 2(-1)^3 - 5(-1) + 4 = -2 + 5 + 4 = 7$. $f(2) = 2(2)^3 - 5(2) + 4 = 16 - 10 + 4 = 10$. Step 2: Compute the difference quotient: $\text{AROC} = \frac{f(2) - f(-1)}{2 - (-1)} = \frac{10 - 7}{3} = \frac{3}{3} = \mathbf{1}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 4.01 Rational Expressions**

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 5.01 Exponential & Logarithmic Functions**

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 6.01 Radicals & Rational Exponents**

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi + 2\pi k}{4}\right) + i\sin\left(\frac{\pi + 2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right]$, $k \in \{0, 1, 2, 3\}$. Step 3: Evaluate for each k:$k = 0$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(2)$.

a. 19 ✓

b. 24

c. 10

d. 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(2)$.

a. 15 ✓

b. 21

c. 12

d. 19

Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for notation for sequences and series consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for arithmetics sequences consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for arithmetic series consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove: in any geometric sequence with positive terms, the geometric mean of two terms equals the intermediate terms.

a. Not always true

b. Only for $r = 1$

c. Use AM-GM

d. **If a, ar, ar^2: $\sqrt{a \cdot ar^2} = ar =$ middle term** ✓

$\sqrt{a \cdot ar^2} = a\sqrt{r^2} = ar$. ✓.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

a. $\frac{5}{4}$ ✓

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

1. **Determine the relation between d and a_1:** The terms of the arithmetic sequence are $a_n = a_1 + (n-1)d$. Therefore: $g_1 = a_1, \quad g_2 = a_3 = a_1 + 2d, \quad g_3 = a_4 = a_1 + 3d$ Since g_1, g_2, g_3 form a geometric progression, the geometric mean property requires: $g_2^2 = g_1 g_3 \Rightarrow (a_1 + 2d)^2 = a_1(a_1 + 3d)$ Expanding both sides: $a_1^2 + 4a_1 d + 4d^2 = a_1^2 + 3a_1 d \; a_1 d + 4d^2 = 0 \Rightarrow d(a_1 + 4d) = 0$ Since $de0$, we have $a_1 + 4d = 0$, which gives.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider an infinite geometric series with first term a and common ratio r, where $|r| < 1$: $a + ar + ar^2 + ar^3 + \ldots = 12$ If each term in the series is squared, the sum of the resulting infinite geometric series is: $a^2 + a^2r^2 + a^2r^4 + a^2r^6 + \ldots = 72$ What is the value of the common ratio, r?

a. $\frac{1}{3}$ ✓

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

Step 1: Use the sum formula for an infinite geometric series with $|r| < 1$: $S = \frac{a}{1-r} = 12 \Rightarrow a = 12(1-r)$ Step 2: When each term is squared, the new series has first term a^2 and common ratio r^2. Since $|r| < 1$, we also have $r^2 < 1$. Its sum is: $S_{\text{squared}} = \frac{a^2}{1-r^2} = 72$ Step 3: Factor the denominator using the difference of squares, $1 - r^2 = (1-r)(1+r)$: $\frac{a^2}{(1-r)(1+r)} = (\frac{a}{1-r})(\frac{a}{1+r}) = 72$ Substi.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. **3** ✓

c. 7

d. 9

Let the geometric sequence be $a_1, a_1r, a_1r^2, \ldots$. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a_1(1-r^n)}{1-r}$ (for $req1$). We are given that the sum of the first 3 terms is 21: $S_3 = \frac{a_1(1-r^3)}{1-r} = 21$ (Equation 1) We are also given that the sum of the first 6 terms is 189: $S_6 = \frac{a_1(1-r^6)}{1-r} = 189$ (Equation 2) We can rewrite Equation 2 using the difference of squares factorization, $1 - r^6 = (1-r^3)(1+r^3)$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic sequence begins with the terms: $8, 13, 18, 23, \ldots$ What is the 15th term of the sequence?

a. 78 ✓

b. 83

c. 73

d. 117

To find the 15th term of an arithmetic sequence, use the explicit formula for the nth term: $a_n = a_1 + (n-1)d$ 1. Identify the first term: $a_1 = 8$. 2. Find the common difference d by subtracting consecutive terms: $d = 13 - 8 = 5$ 3. Substitute $n = 15$, $a_1 = 8$, and $d = 5$ into the formula: $a_{15} = 8 + (15-1)(5) = 8 + 14(5) = 8 + 70 = 78$ Common Misconceptions: - Choice B (83) results from forgetting to subtract 1 from n, computing $8 + 15(5) = 83$. - Choice C (73) results from off-by-one in.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the arithmetic sequence where the first term $a_1 = 7$ and the common difference $d = -3$. Which of the following is the explicit formula for the nth term of this sequence?

a. $a_n = 7 + 3(n-1)$

b. $a_n = 7 - 3(n-1)$ ✓

c. $a_n = 3 - 7(n-1)$

d. $a_n = -3 + 7(n-1)$

An arithmetic sequence is defined by the formula $a_n = a_1 + (n-1)d$, where a_n is the nth term, a_1 is the first term, and d is the common difference. In this problem, we are given that the first term $a_1 = 7$ and the common difference $d = -3$. Substituting these values into the formula, we get $a_n = 7 + (n-1)(-3)$, which can be rewritten as $a_n = 7 - 3(n-1)$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A sequence is defined by the recurrence relation $a_n = 2a_{n-1} - a_{n-2} + 3$ for $n \geq 3$. If the first term $a_1 = 5$ and the second term $a_2 = 8$, find the value of a_4.

a. 17

b. 20

c. **23** ✓

d. 26

The recurrence relation is given by $a_n = 2a_{n-1} - a_{n-2} + 3$ for $n \geq 3$. We are given $a_1 = 5$ and $a_2 = 8$. To find a_3, we set $n = 3$: $a_3 = 2a_2 - a_1 + 3$ $a_3 = 2(8) - 5 + 3$ $a_3 = 16 - 5 + 3$ $a_3 = 11 + 3$ $a_3 = 14$ To find a_4, we set $n = 4$: $a_4 = 2a_3 - a_2 + 3$ $a_4 = 2(14) - 8 + 3$ $a_4 = 28 - 8 + 3$ $a_4 = 20 + 3$ $a_4 = 23$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $|r| < 1$, prove $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ using the limit of partial sums.

a. Use calculus only

b. Cannot be proved

c. $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ **as** $r^{N+1} \to 0$ ✓

d. Only for $0 < r < 1$

Applying Sequences & Series principles, Classic geometric series proof. Therefore, the verified correct option is $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ as $r^{N+1} \to 0$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. **By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation** ✓

d. Use telescoping

Applying Sequences & Series principles, Standard induction. Therefore, the verified correct option is By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find a_1 and d for the arithmetic sequence where $S_{10} = 155$ and $a_{10} = 28$.

a. $a_1 = 3,\ d = \frac{25}{9}$ ✓

b. $a_1 = 5,\ d = 3$

c. $a_1 = 1,\ d = 3$

d. $a_1 = 3,\ d = 3$

From $S_{10} = \frac{10}{2}(a_1 + a_{10}) = 155$ with $a_{10} = 28$, we get $5(a_1 + 28) = 155 \Rightarrow a_1 = 3$. Then $a_{10} = a_1 + 9d \Rightarrow 28 = 3 + 9d \Rightarrow d = \frac{25}{9}$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Prove by induction: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

a. Not provable

b. Only for even n

c. Use calculus

d. **Base: $n = 1$: $1 = 6/6$. ✓. Step: add $(n+1)^2$ and factor. ✓**

Applying Sequences & Series principles, Standard induction. Therefore, the verified correct option is Base: $n = 1$: $1 = 6/6$. ✓. Step: add $(n+1)^2$ and factor..

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the closed form for F_n (Fibonacci) using Binet's formula.

a. $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ **where** $= 1 + 52, = 1 - 52$ ✓

b. $F_n = \phi^{n-1}$

c. $F_n = 2^n/5$

d. $F_n = n\phi$

Applying Sequences & Series principles, Binet's formula. Therefore, the verified correct option is $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ where $= 1 + 52, = 1 - 52$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rational Expressions

Using partial fractions, find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$.

a. 1

b. $\frac{1}{2}$

c. $\frac{n}{2n-1}$

d. $\frac{n}{2n+1}$ ✓

PFD: $\frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$. Telescopes to $\frac{1}{2}\left(1 - \frac{1}{2n+1}\right) = \frac{n}{2n+1}$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The formula for the sum of n terms of an arithmetic series is:

a. $S_n = \frac{a_1}{1-r}$

b. $S_n = \frac{n}{2}(a_1 + a_n)$ ✓

c. $S_n = na_1$

d. $S_n = a_1 \cdot \frac{r^n - 1}{r - 1}$

Applying Sequences & Series principles, Arithmetic sum formula. Therefore, the verified correct option is $S_n = \frac{n}{2}(a_1 + a_n)$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Radicals & Rational Exponents

A geometric series with $|r| < 1$ has an infinite sum of:

a. $S = a_1 \cdot r$

b. $S = \frac{a_1}{1-r}$ ✓

c. Undefined

d. $S = \frac{a_1}{r}$

Applying Sequences & Series principles, Infinite geometric series sum. Therefore, the verified correct option is $S = \frac{a_1}{1-r}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

Find the first term and common difference of the AP whose pth term is $3p + 2$.

a. $a_1 = 2, d = 3$

b. $a_1 = 5, d = 3$ ✓

c. $a_1 = 5, d = 2$

d. $a_1 = 3, d = 2$

Applying Sequences & Series principles, $a_p = 3p + 2 \Rightarrow a_1 = 5, d = 3$. Therefore, the verified correct option is $a_1 = 5, d = 3$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Trigonometric Functions

Use sigma notation for $1^2 + 2^2 + 3^2 + \cdots + n^2$.

a. $\prod_{k=1}^{n} k^2$

b. $\sum_{k=0}^{n} k^2$

c. $\sum_{k=1}^{n} k^2$ ✓

d. n^2

Applying Sequences & Series principles, Standard sigma notation. Therefore, the verified correct option is $\sum_{k=1}^{n} k^2$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Functions & Their Graphs

Insert 3 arithmetic means between 5 and 21.

a. $7, 11, 15$

b. $9, 13, 17$ ✓

c. $8, 12, 16$

d. $10, 14, 18$

Applying Sequences & Series principles, $d = (21 - 5)/4 = 4$. Means: $9, 13, 17$. Therefore, the verified correct option is $9, 13, 17$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Matrices & Systems

Is the series $1 - 1 + 1 - 1 + \cdots$ convergent?

a. Yes, sum = 1

b. Yes, sum = 0

c. **No (it diverges)** ✓

d. Yes, sum = 1/2

Applying Sequences & Series principles, Alternating 1s: no limit. Therefore, the verified correct option is No (it diverges).

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability & Statistics

The sum of the first n squares: $\sum_{k=1}^{n} k^2 =$

a. $\frac{n(n+1)}{2}$

b. n^3

c. $\frac{n^2(n+1)}{2}$

d. $\frac{n(n+1)(2n+1)}{6}$ ✓

Applying Sequences & Series principles, Standard formula. Therefore, the verified correct option is $\frac{n(n+1)(2n+1)}{6}$.

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